

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

- | | |
|-------------------------------------|--|
| n/a | Confirmed |
| ☐ | ☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☐ | ☒ A description of all covariates tested |
| ☐ | ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	LEGSys device BLE data was collected using LEGSys software v3.0.5 (Biosensics LLC). Biosymbiotic device BLE data was collected using custom Python code written using Python 3.10. Device power characterization data was collected using STM32CubeMonitor-Power software v1.2.1 (STMicroelectronics NV). Python 3.10 packages utilized for data collection: bleak==0.22.3, bleak-winrt==1.2.0, matplotlib==3.10.3, numpy==2.2.5, pandas==2.2.3.
Data analysis	All data were analyzed using custom code written in Python 3.10 or MATLAB R2024b. Python 3.10 packages used for data analysis: keras==3.11.3, matplotlib==3.10.3, numpy==2.2.5, onnx==1.17.0, pandas==2.2.3, scikit-learn==1.6.1, skl2onnx==1.18.0, sktime==0.37.0, tensorflow==2.20.0, seaborn==0.13.2.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All data supporting the findings of this study are available within the article and its supplementary files. Any additional requests for information can be directed to, and will be fulfilled by, the corresponding author. Datasets used to generate the figures (with relevant code) are available on ReData (<https://redata.arizona.edu/>), the University of Arizona's official public data archive. DOI: 10.25422/azu.data.29614193

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	Gender data were not collected. Self-reported sex data were collected from demographic questionnaires during gold standard comparison and in vivo device validation, but used only for frailty status determination. Sex- and gender-based analyses were not used in this study, as each participant served as their own control to eliminate inter-subject physiological variability. Furthermore, the limited sample sizes would not provide meaningful data for sufficient subgroup analysis.
Reporting on race, ethnicity, or other socially relevant groupings	No data regarding race, ethnicity, or social groupings were collected during this study. These data are not relevant for this proof-of-concept device study.
Population characteristics	Subjects aged 65 and older with no history of sensorimotor disorders or conditions (Parkinson's disease, multiple sclerosis, stroke, recent hip, leg, or foot surgery, etc). Subjects must also have been able to walk 60 seconds unassisted, able to achieve a Mini-Mental Examination score greater than 23, had no terminal illnesses, had no preexisting skin disease, and must have been able to speak, read, and write English.
Recruitment	Subjects were recruited using flyers posted at the university medical center (Banner University Medical Center), through phone calls, face to face communication, and from the previously concluded NIH-funded Arizona Frailty Cohort Study (1412591150). There is likely bias for subjects recruited via posted flyer or face-to-face, as these subjects would necessarily have some degree of mobility and independence. There is also bias introduced by requiring subjects to travel independently to the designated research location at the university medical center. These approaches self-select for subjects who are not acutely frail, which limited recruitment and sample size.
Ethics oversight	All studies were approved by a University of Arizona Internal Review Board, protocol #2005648028 for in vivo studies and protocol #STUDY00005692 for ten-day continuous wear trials.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample size for gold-standard device comparison N1 = 16. Sample size for in vivo device validation N2 = 14. No sample-size calculations were performed, sample sizes were restricted by the limited cohort of pre-frail / frail subjects in the study area and the ability of these subjects to travel to the IRB-approved research site.
Data exclusions	Data from 4 original cohort subjects were not included in the main figures nor during model training as these subjects did not complete the grip strength evaluation and their frailty status (and therefore their label for model training) could not be determined. These criteria were pre-established.
Replication	Each participant served as their own control with simultaneous measurements from both the experimental and gold-standard devices on the same placements during the same gait session. This inherently verified reproducibility at the individual level by demonstrating consistent agreement between devices across all subjects. Reproducibility was assessed across multiple derived gait metrics, with unpaired t-tests showing high P-values (mean = 0.81) and negligible effect sizes (Cohen's d mean: 0.08), indicating consistent device performance across different measurement parameters. The 10-day in vivo study validated device functionality was reproducible in various environments over extended durations with continuous, unsupervised use. Note that the reproducibility focus was measurement fidelity, rather than reproducibility of individual gait patterns, which vary highly within and between subjects. All validation attempts were successful.

Randomization	Randomization was not relevant for this study, each subject served as their control for gold standard validation with simultaneous measurements from both the experimental and gold-standard devices on the same placements during the same gait session. This experimental design eliminated the need for group allocation, i.e., comparing devices within the same individual accounted for any physiological variability (age, sex, fitness, gait patterns, etc.) between subjects.
Blinding	Blinding was also not relevant for this study, there were no experimental groups, subjective assessments, nor interventions to blind investigators against. Data collection relied on automated angular velocity data recorded by a digital sensor. The angular velocity data are objective and are not subject to observer interpretation or influence.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	<i>Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.</i>
Novel plant genotypes	<i>Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.</i>
Authentication	<i>Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.</i>